



Prevention Without Proof

Did national suicide-prevention strategies reduce suicide?

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40s

Somewhere on earth, about every forty seconds, a person decides the rest of life is not worth the rest of life.



More than war, more than homicide

More than **700,000** suicides a year.

More than war.

More than homicide.

More than most diseases we built ministries
to defeat.

The number is stubborn

It has barely moved in a generation.



The world answered with a document

Finland, 1986–92: investigate every suicide, build a national program.

Then everyone copied it: Norway and Sweden 1995, France 2000, the UK 2002, Ireland 2005.

By 2018 the WHO urges *every* member state to adopt one.

The premise

We decided, collectively, that the **strategy document** is the cure.

The only question that matters

Does a country that adopts a national strategy
bury **fewer** of its citizens than it otherwise would have?

Empirical. Answerable in principle. Almost entirely unanswered in practice.



My answer, stated up front

From the cross-national record, **we cannot tell** whether these strategies have saved a single life.

That should frighten the skeptic and the believer alike.

The famous answer

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And the method underneath it



The canonical finding looked protective

Matsubayashi & Ueda (2011): across 21 OECD countries, those with national programs saw suicide **fall** relative to those without.

It became the evidence base for a global policy.

The engine

A two-way fixed-effects comparison of adopters vs. non-adopters.



Two-way fixed effects misleads under staggered timing

When units adopt at different dates and effects are heterogeneous, TWFE uses **already-treated units as controls** for newly-treated ones.

- The weights can be negative.
- The estimate can fall outside the range of every true effect.
- It can return the **wrong sign** for mechanical reasons.

Goodman-Bacon (2021); de Chaisemartin & D'Haultfœuille (2020).

The data

All real — nothing simulated

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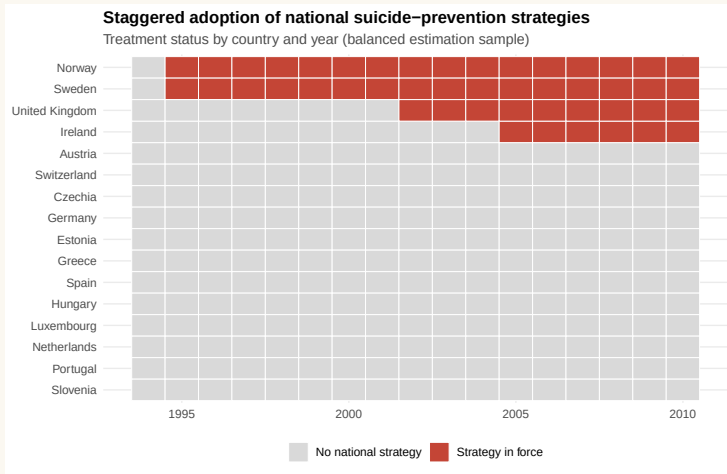
Three real sources, one country-year panel

- **Outcome:** Eurostat age-standardised suicide rate (ICD-10 X60–X84, Y87.0), 1994–2010.
- **Treatment:** year a government adopted a stand-alone national strategy (WHO 2018; Lewitzka et al. 2019).
- **Covariate:** World Bank unemployment — the business cycle.

Verified independently in R, Python, and Stata: identical samples and means.

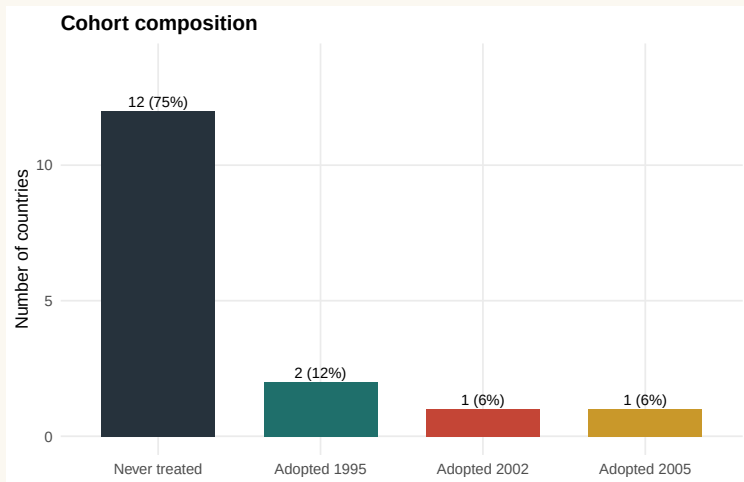


The natural experiment is staggered — and thin



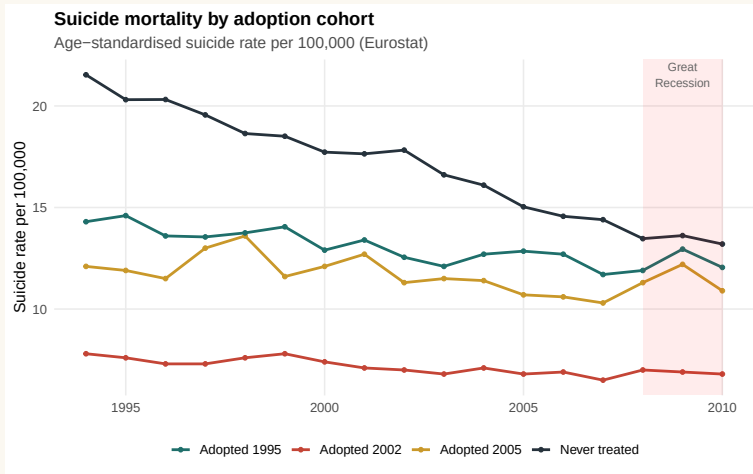


Four treated countries — I will not pretend otherwise





Treated countries start lower; all bend up in the recession



The design

What we are trying to measure



The estimand: a weighted average effect on the treated

For each adoption cohort g and year t , compare the change in suicide against not-yet- and never-treated countries; aggregate, weighting cohorts by size.

$$\theta = \sum_g w_g \overline{ATT}(g), \quad w_g \propto \# \text{treated countries in } g.$$

Callaway & Sant'Anna (2021), under *conditional* parallel trends.



The covariate an expert would insist on

Recessions raise suicide. Recoveries lower it.

And Europe's recessions are not independent of its strategies.

So we condition on **unemployment** — the fast-moving confounder that can wear a policy's clothes (Ruhm 2000; Reeves et al. 2012).



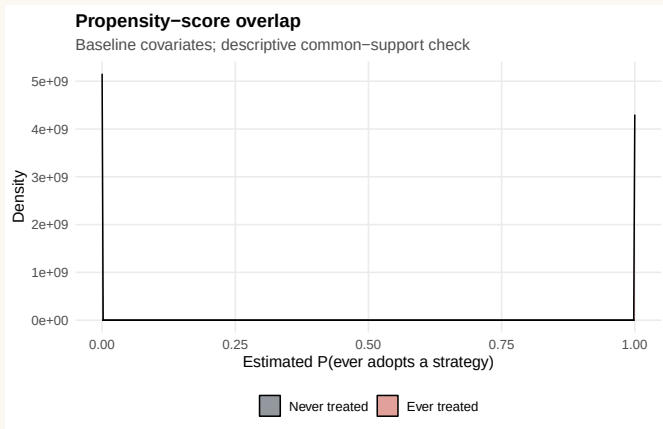
And unemployment is genuinely imbalanced

Baseline (1994)	Treated	Control	Std. diff.
Suicide rate (per 100k)	12.1	21.5	-0.99
Unemployment (%)	9.8	8.2	+0.33
Share aged 65+ (%)	15.2	14.0	+0.63
Median age (years)	35.2	36.2	-0.42

Imbens & Rubin (2015): $|\text{std. diff}| > 0.25$ is imbalanced. Unemployment clears it — conditioning is necessary, not cosmetic.



There is no common support to weight on



Perfect separation $\Rightarrow$ regression adjustment, not inverse-propensity weighting.



Sixteen clusters: trust the placebo, not the t -statistic

Four treated countries, twelve controls $\approx$ sixteen clusters.

Cluster-robust and bootstrap standard errors need many-cluster asymptotics this design does not have.

So the inferential weight is carried by randomization inference

How often does *fake* adoption, assigned at random among the never-treated, produce an estimate this large?

What the data say

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The inconvenient sign



The estimate is positive: more suicide after adoption

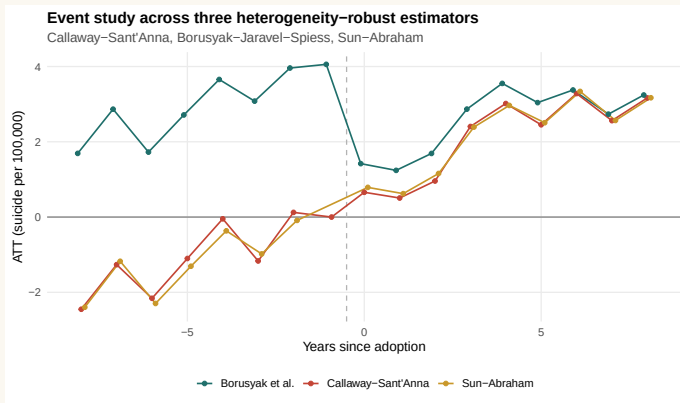
+2.7

suicides per 100,000, after adoption (s.e. 1.05)

Group-size-weighted ATT, Callaway–Sant’Anna, conditioning on unemployment.



Three heterogeneity-robust estimators agree



Callaway-Sant'Anna +2.7 · Borusyak et al. +3.2 · Sun-Abraham +3.0.

A careless reader would stop here

“Suicide-prevention strategies are followed by *more* suicide.”

Do not stop here.

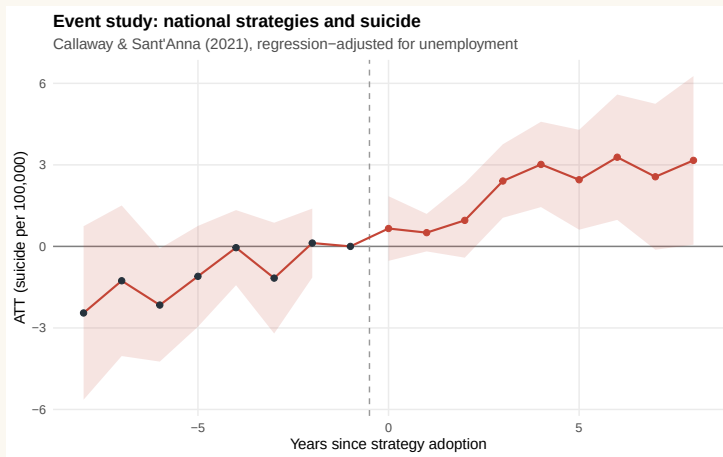
What the number is *not*

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Five windows, one answer



Measured correctly, pre-trends are flat



Long differences from $g-1$: $\sum z^2 = 9.9$ (8 d.f.) — parallel trends not rejected.



A baseline can manufacture a pre-trend

Short differences (varying base)

Each pre-coefficient is a one-period change.

$$\sum z^2 = 59 \Rightarrow \text{"trends fail."}$$

Long differences (fixed $g-1$)

Cumulative deviation from $g-1$ — the same baseline as every post-coefficient, and as parallel trends itself.

$$\sum z^2 = 9.9 \Rightarrow \text{flat.}$$

The “violation” was an artifact of the wrong baseline. Most applied work reports the wrong one.

But flat is not a clean bill of health

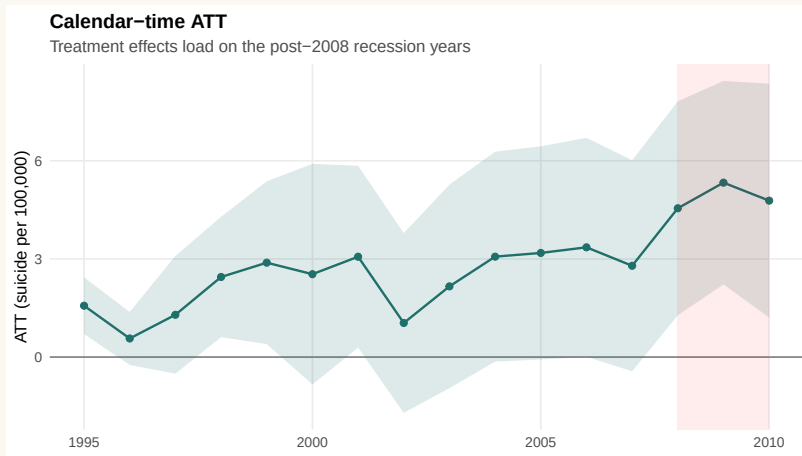
The pre-trend leads are identified almost entirely off **the UK and Ireland**; Norway and Sweden (one pre-period each) feed the *level* of the effect but not the *test* of trends.

With four treated countries the test catches only gross violations.

It catches none — the silence of a design too small to hear anything, not vindication.



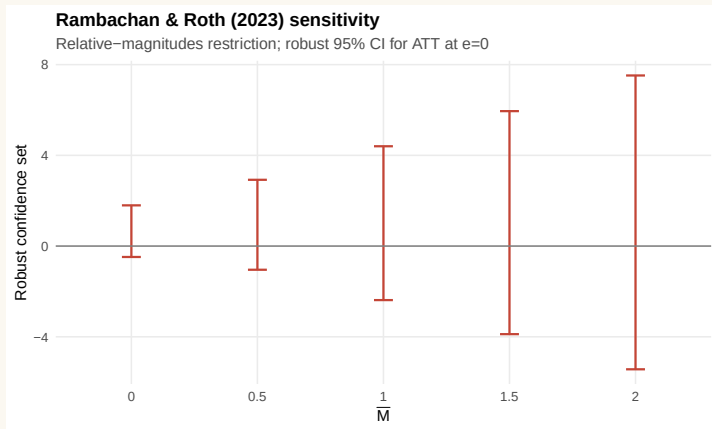
The “effect” loads on the 2008 recession



The later cohorts' post-periods overlap Europe's austerity years.

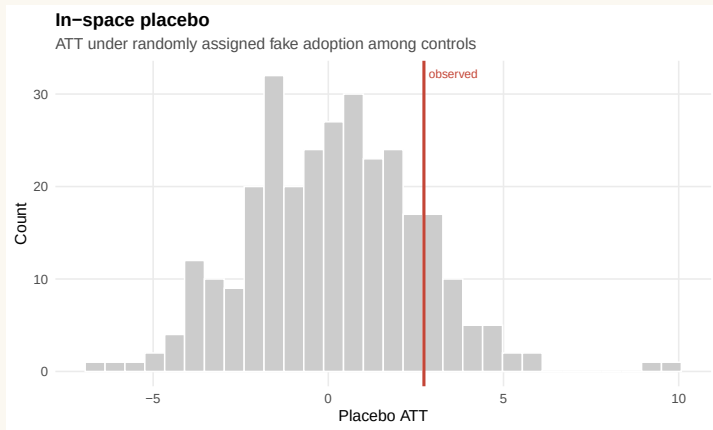


The effect cannot be signed



Rambachan & Roth (2023): under modest trend violations the robust CI spans zero.

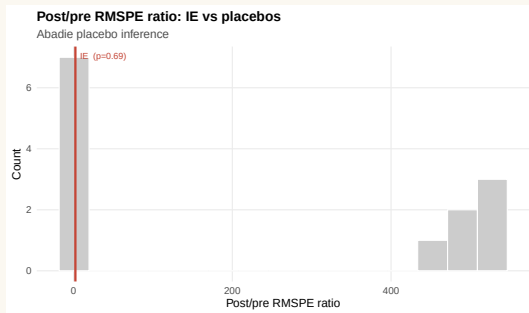
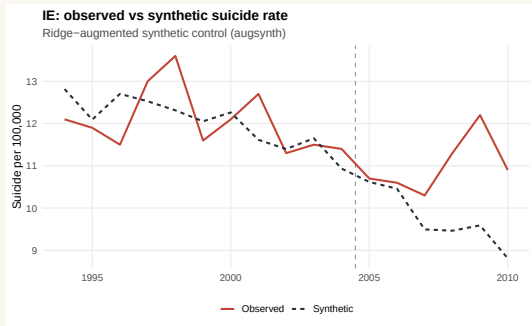
Indistinguishable from chance



One placebo in four, drawn from never-treated countries, is at least this large ($p = 0.25$).



Even the method of last resort says nothing



Augmented synthetic control: placebo $p = 0.46$ (UK), 0.69 (Ireland). Ireland's gap opens in 2008.



Fragile where a real effect would be solid

Drop the four post-communist controls,
and the estimate nearly halves: $+2.7 \rightarrow +1.2$.

A real treatment effect should not depend on which untreated countries you compare against. This one does.



Is it just bad two-way fixed effects? No.

Goodman-Bacon 2×2	Weight	Est.	Contrib.
Treated vs. untreated	0.86	3.57	3.085
Later vs. earlier treated	0.11	0.29	0.032
Earlier vs. later treated	0.03	-0.22	-0.005
Reconstructed TWFE	1.00		3.11

The “forbidden” comparisons carry $\sim 3\%$ weight. TWFE (3.11) $\approx$ CS (2.7). The estimator is not the problem — identification is.

Devil's advocate

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The strongest objection



“You are explaining away an inconvenient result”

The skeptic

You got a positive estimate and then argued it away with pre-trends and a recession. Convenient.

The response

I am *not* claiming the recession explains the result — that would identify, from a panel I called uninformative, an effect I have no business identifying. The claim is weaker and more durable: these forces are present, the design cannot separate them, and a number that confounds all three measures none.

What this means

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For the question, and for the Commission



Not that strategies fail — that we cannot see

The finding is not that prevention is worthless.
It is that the observational cross-section is **structurally incapable**
of revealing its worth.

A policy can be both worth keeping and unproven. Honesty means holding both.



We built a continent of strategies with no evaluation inside

No staggered randomization. No phased regional rollout with administrative mortality data. No pre-registered evaluation attached to a single national plan.

Every country switched at once, at the level where the confounders live, in response to the outcome itself.



For the Commission: design the evaluation before the policy

- Build the counterfactual **in**, not reconstruct it afterward.
- Evaluate **below the nation** — regions, phased rollout.
- Pre-register the mortality outcome before deployment.

The alternative is to keep spending moral and fiscal capital on faith.

Italy never wrote the document

The country in which we sit is a documented **never-adopter**.

A large European nation that never wrote the strategy —
the silent control against which the writers might have been judged,
had the data let us judge them.

We have been making life-and-death policy

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in a darkness we mistook for evidence.